

## KEKO v4: Uncertainty as a Reinforcement Substrate for Self-Adaptive Neural Systems

**Abstract:** This paper presents KEKO v4, an uncertainty-driven self-adaptive neural framework that transforms epistemic uncertainty into a reinforcement substrate for continual learning. Unlike conventional confidence-based models, KEKO leverages uncertainty as both a diagnostic and developmental signal. The architecture automatically detects, labels, and ingests uncertain tokens, using their recurrence and resolution to drive autonomous optimization.

- 1. Introduction** Modern neural architectures traditionally minimize uncertainty as noise. However, uncertainty is also a rich epistemic signal: it encodes what the model does not yet know and where it can grow. The KEKO framework reinterprets uncertainty not as error, but as potential energy for learning.
- 2. Related Work** Prior research includes Bayesian epistemic uncertainty estimation, active learning strategies, and meta-cognitive control systems. KEKO unifies these approaches under a distributed reinforcement paradigm.
- 3. System Architecture** The architecture is composed of four layers: Recognition, Clarification, Reinforcement, and Meta-Cognitive supervision. Each layer contributes to a continuous epistemic feedback loop.
- 4. Experimental Framework** Key metrics include AUT (Average Uncertainty per Token), URR (Uncertainty Reduction Rate), TSM (Temporal Stability Metric), and  $\Delta U$  distributions. Early findings show reduced confidence drift and improved specialization stability.
- 5. Discussion and Conclusion** KEKO v4 demonstrates that uncertainty can function as a constructive feedback substrate for autonomous learning. Future work includes extending to multimodal domains, tracing uncertainty diffusion, and evaluating uncertainty-topology correlations.